

```
7  -- 1. DATA EXPLORATION & CLEANING
8  -----
```

```
10 -- Preview the data
11 SELECT * FROM "Sample - Superstore" LIMIT 10;
```

```
13 -- Check total row count
14 SELECT COUNT(*) AS Total_Rows FROM "Sample - Superstore";
```

```
16 -- Check for null values in key columns
17 SELECT
18     SUM(CASE WHEN Sales IS NULL THEN 1 ELSE 0 END) AS Null_Sales,
```

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID
1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10001798
2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10000454
3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240
4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10000577
5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10000760
6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10001487
7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10002833
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002275
9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-BI-10003910
10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-10002892

```

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15
16 -- Check for null values in key columns
17 SELECT
18     SUM(CASE WHEN Sales IS NULL THEN 1 ELSE 0 END) AS Null_Sales,

```

	Total_Rows
1	9994

```

14 SELECT COUNT(*) AS Total_Rows FROM "Sample - Superstore";
15
16 -- Check for null values in key columns
17 SELECT
18     SUM(CASE WHEN Sales IS NULL THEN 1 ELSE 0 END) AS Null_Sales,
19     SUM(CASE WHEN Profit IS NULL THEN 1 ELSE 0 END) AS Null_Profit,
20     SUM(CASE WHEN Category IS NULL THEN 1 ELSE 0 END) AS Null_Category,
21     SUM(CASE WHEN "Sub-Category" IS NULL THEN 1 ELSE 0 END) AS Null_SubCategory,
22     SUM(CASE WHEN Region IS NULL THEN 1 ELSE 0 END) AS Null_Region
23 FROM "Sample - Superstore";
24

```

	Null_Sales	Null_Profit	Null_Category	Null_SubCategory	Null_Region
1	0	0	0	0	0

```

23 FROM "Sample - Superstore";
24
25 -- Check for duplicate line items (same order + product appearing more than once)
26 SELECT "Order ID", "Product ID", COUNT(*) AS Cnt
27 FROM "Sample - Superstore"
28 GROUP BY "Order ID", "Product ID"
29 HAVING COUNT(*) > 1;
30
31 -- Create a cleaned table (drops exact duplicates + rows missing core numeric fields)
32 -- Use this table for every query below
33 CREATE TABLE superstore_clean AS

```

	Order ID	Product ID	Cnt
1	CA-2015-103135	OFF-BI-10000069	2
2	CA-2016-129714	OFF-PA-10001970	2
3	CA-2016-137043	FUR-FU-10003664	2
4	CA-2016-140571	OFF-PA-10001954	2
5	CA-2017-118017	TEC-AC-10002006	2
6	CA-2017-152912	OFF-ST-10003208	2
7	US-2014-150119	FUR-CH-10002965	2
8	US-2016-123750	TEC-AC-10004659	2

```
30
31 -- Create a cleaned table (drops exact duplicates + rows missing core numeric fields)
32 -- Use this table for every query below
33 CREATE TABLE superstore_clean AS
34 SELECT DISTINCT * FROM "Sample - Superstore"
35 WHERE Sales IS NOT NULL AND Profit IS NOT NULL;
36
37 -- =====
38 -- 2. PROFIT MARGIN ANALYSIS
39 -- =====
```

```

41 -- Profit margin by Category
42 SELECT Category,
43        ROUND(SUM(Sales), 2) AS Total_Sales,
44        ROUND(SUM(Profit), 2) AS Total_Profit,
45        ROUND(SUM(Profit) * 100.0 / SUM(Sales), 2) AS Profit_Margin_Pct
46 FROM superstore_clean
47 GROUP BY Category
48 ORDER BY Profit_Margin_Pct DESC;
49
50 -- Profit margin by Sub-Category (loss-makers float to top)
51 SELECT "Sub-Category",

```

	Category	Total_Sales	Total_Profit	Profit_Margin_Pct
1	Technology	836154.03	145454.95	17.4
2	Office Supplies	719047.03	122490.8	17.04
3	Furniture	741999.8	18451.27	2.49

```

49
50 -- Profit margin by Sub-Category (loss-makers float to top)
51 SELECT "Sub-Category",
52        ROUND(SUM(Sales), 2) AS Total_Sales,
53        ROUND(SUM(Profit), 2) AS Total_Profit,
54        ROUND(SUM(Profit) * 100.0 / SUM(Sales), 2) AS Profit_Margin_Pct
55 FROM superstore_clean
56 GROUP BY "Sub-Category"
57 ORDER BY Profit_Margin_Pct ASC;
58
59 -- Top 5 most profitable sub-categories
60 SELECT "Sub-Category", ROUND(SUM(Profit), 2) AS Total_Profit

```

	Sub-Category	Total_Sales	Total_Profit	Profit_Margin_Pct
1	Tables	206965.53	-17725.48	-8.56
2	Bookcases	114880.0	-3472.56	-3.02
3	Supplies	46673.54	-1189.1	-2.55
4	Machines	189238.63	3384.76	1.79
5	Chairs	328449.1	26590.17	8.1
6	Storage	223843.61	21278.83	9.51
7	Phones	330007.05	44515.73	13.49
8	Furnishings	91705.16	13059.14	14.24
9	Binders	203412.73	30221.76	14.86
10	Appliances	107532.16	18138.01	16.87
11	Art	27118.79	6527.79	24.07

```

57 ORDER BY Profit_margin_F00 ASC,
58
59 -- Top 5 most profitable sub-categories
60 SELECT "Sub-Category", ROUND(SUM(Profit), 2) AS Total_Profit
61 FROM superstore_clean
62 GROUP BY "Sub-Category"
63 ORDER BY Total_Profit DESC
64 LIMIT 5;
65
66 -- Bottom 5 loss-making sub-categories

```

	Sub-Category	Total_Profit
1	Copiers	55617.82
2	Phones	44515.73
3	Accessories	41936.64
4	Paper	34053.57
5	Binders	30221.76

```
65
66 -- Bottom 5 loss-making sub-categories
67 SELECT "Sub-Category", ROUND(SUM(Profit), 2) AS Total_Profit
68 FROM superstore_clean
69 GROUP BY "Sub-Category"
70 ORDER BY Total_Profit ASC
71 LIMIT 5;
72
73 -- =====
74 -- 3. REGIONAL ANALYSIS
75 -- =====
```

	Sub-Category	Total_Profit
1	Tables	-17725.48
2	Bookcases	-3472.56
3	Supplies	-1189.1
4	Fasteners	949.52
5	Machines	3384.76

```

77 -- Profit and sales by Region
78 SELECT Region,
79        ROUND(SUM(Sales), 2) AS Total_Sales,
80        ROUND(SUM(Profit), 2) AS Total_Profit,
81        ROUND(SUM(Profit) * 100.0 / SUM(Sales), 2) AS Profit_Margin_Pct
82 FROM superstore_clean
83 GROUP BY Region
84 ORDER BY Total_Profit DESC;
85
86 -- Region + Category drill-down
87 SELECT Region, Category,

```

	Region	Total_Sales	Total_Profit	Profit_Margin_Pct
1	West	725457.82	108418.45	14.94
2	East	678781.24	91522.78	13.48
3	South	391721.91	46749.43	11.93
4	Central	501239.89	39706.36	7.92

```

85
86 -- Region + Category drill-down
87 SELECT Region, Category,
88        ROUND(SUM(Sales), 2) AS Total_Sales,
89        ROUND(SUM(Profit), 2) AS Total_Profit
90 FROM superstore_clean
91 GROUP BY Region, Category
92 ORDER BY Region, Total_Profit DESC;
93

```

	Region	Category	Total_Sales	Total_Profit
1	Central	Technology	170416.31	33697.43
2	Central	Office Supplies	167026.42	8879.98
3	Central	Furniture	163797.16	-2871.05
4	East	Technology	264973.98	47462.04
5	East	Office Supplies	205516.05	41014.58
6	East	Furniture	208291.2	3046.17
7	South	Technology	148771.91	19991.83
8	South	Office Supplies	125651.31	19986.39
9	South	Furniture	117298.68	6771.21
10	West	Office Supplies	220853.25	52609.85
11	West	Technology	251991.83	44303.65

```

97
98 -- Average discount vs total profit by Category
99 SELECT Category,
100         ROUND(AVG(Discount), 3) AS Avg_Discount,
101         ROUND(SUM(Profit), 2) AS Total_Profit
102 FROM superstore_clean
103 GROUP BY Category;
104
105 -- Discount bands vs average profit (shows the tipping point where discounts erase profit)

```

	Category	Avg_Discount	Total_Profit
1	Furniture	0.174	18451.27
2	Office Supplies	0.157	122490.8
3	Technology	0.132	145454.95

```

105  -- Discount bands vs average profit (shows the tipping point where discounts erase profit)
106  SELECT
107  CASE
108      WHEN Discount = 0 THEN '0%'
109      WHEN Discount <= 0.2 THEN '1-20%'
110      WHEN Discount <= 0.4 THEN '21-40%'
111      ELSE '40%+'
112  END AS Discount_Band,
113  ROUND(AVG(Profit), 2) AS Avg_Profit,
114  COUNT(*) AS Order_Count
115  FROM superstore_clean
116  GROUP BY Discount_Band
117  ORDER BY Discount_Band;
118
119

```

	Discount_Band	Avg_Profit	Order_Count
1	0%	66.9	4798
2	1-20%	26.5	3803
3	21-40%	-77.86	460
4	40%+	-106.71	933

```

123 -- Monthly sales and profit trend
124 SELECT strftime('%Y-%m', "Order Date") AS Order_Month,
125         ROUND(SUM(Sales), 2) AS Total_Sales,
126         ROUND(SUM(Profit), 2) AS Total_Profit
127 FROM superstore_clean
128 GROUP BY Order_Month
129 ORDER BY Order_Month;
130
131 -- Sales by quarter and category (seasonal pattern)
132 SELECT
133     CASE
134         WHEN CAST(strftime('%m', "Order Date") AS INTEGER) IN (1, 2, 3) THEN 'Q1'

```

	Order_Month	Total_Sales	Total_Profit
1	NULL	2297200.86	286397.02

```

130
131 -- Sales by quarter and category (seasonal pattern)
132 SELECT
133 CASE
134     WHEN CAST(strftime('%m', "Order Date") AS INTEGER) IN (1,2,3) THEN 'Q1'
135     WHEN CAST(strftime('%m', "Order Date") AS INTEGER) IN (4,5,6) THEN 'Q2'
136     WHEN CAST(strftime('%m', "Order Date") AS INTEGER) IN (7,8,9) THEN 'Q3'
137     ELSE 'Q4'
138 END AS Quarter,
139 Category,
140 ROUND(SUM(Sales), 2) AS Total_Sales
141 FROM superstore_clean
142 GROUP BY Quarter, Category
143 ORDER BY Quarter, Category;
144
145 -- =====
146 -- 6 ORDER-TO-SHIP DAYS (fulfillment/inventory proxy => feeds Python correlation step)

```

	Quarter	Category	Total_Sales
1	Q4	Furniture	741999.8
2	Q4	Office Supplies	719047.03
3	Q4	Technology	836154.03

```

145 -- =====
146 -- 6. ORDER-TO-SHIP DAYS (fulfillment/inventory proxy -> feeds Python correlation step)
147 -- =====
148
149 -- Row-level days-to-ship (export this for the Pandas correlation analysis)
150 SELECT "Order ID",
151        Category,
152        "Sub-Category",
153        Profit,
154        Sales,
155        "Ship Date"
156 FROM superstore_clean;
157
158 -- Average ship days by Category alongside profit

```

	Order ID	Category	Sub-Category	Profit	Sales	Ship Date
1	CA-2016-152156	Furniture	Bookcases	41.9136	261.96	11/11/2016
2	CA-2016-152156	Furniture	Chairs	219.582	731.94	11/11/2016
3	CA-2016-138688	Office Supplies	Labels	6.8714	14.62	6/16/2016
4	US-2015-108966	Furniture	Tables	-383.031	957.5775	10/18/2015
5	US-2015-108966	Office Supplies	Storage	2.5164	22.368	10/18/2015
6	CA-2014-115812	Furniture	Furnishings	14.1694	48.86	6/14/2014
7	CA-2014-115812	Office Supplies	Art	1.9656	7.28	6/14/2014
8	CA-2014-115812	Technology	Phones	90.7152	907.152	6/14/2014
9	CA-2014-115812	Office Supplies	Binders	5.7825	18.504	6/14/2014

SQL 1*

```
160 -- 7. SHIP MODE / SEGMENT BREAKDOWN (supporting insights)
161 -- =====
162
163 SELECT "Ship Mode",
164        ROUND(SUM(Profit), 2) AS Total_Profit,
165        ROUND(AVG(Discount), 3) AS Avg_Discount
166 FROM superstore_clean
167 GROUP BY "Ship Mode";
168
169 SELECT Segment,
170        ROUND(SUM(Sales), 2) AS Total_Sales,
171        ROUND(SUM(Profit), 2) AS Total_Profit
172 FROM superstore_clean
173 GROUP BY Segment
174 ORDER BY Total_Profit DESC;
175
```

	Segment	Total_Sales	Total_Profit
1	Consumer	1161401.34	134119.21
2	Corporate	706146.37	91979.13
3	Home Office	429653.15	60298.68